

Disproof of conjecture 00000000745:
the Julia set of $x^2 - 1$ on $\mathbb{Z}_2$ is empty, so its Hausdorff dimension is
0, not 1

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1 The conjecture and the reading

Statement (English, verbatim). Conjecture: The Julia set (repelling points) of $x^2 - 1$ on $\mathbb{Z}_2$ has Hausdorff dimension 1, and its complement is open and dense (2-adic Julia topology). (The Chinese text states the same: the Julia set (set of repelling points) of $x^2 - 1$ on $\mathbb{Z}_2$ has Hausdorff dimension 1 and its complement is a dense open set.)

Reading. The conjecture is the conjunction, for the Julia set $J \subseteq \mathbb{Z}_2$ of $f(x) = x^2 - 1$ acting on the 2-adic integers $\mathbb{Z}_2$ with the 2-adic metric $d(x, y) = |x - y|_2$, of

$$(A) \quad \dim_H J = 1 \quad \text{and} \quad (B) \quad \mathbb{Z}_2 \setminus J \text{ is open and dense.}$$

We work on the space the statement names, $\mathbb{Z}_2$, and with the map it names. We formalize four readings of “the Julia set (repelling points)”:

- (J1) the complement of the Fatou set, where the Fatou set is the set of points having a neighbourhood on which the family of iterates $\{f^n\}_{n \geq 0}$ is equicontinuous;
- (J2) the set of points at which the family $\{f^n\}$ is not equicontinuous;
- (J3) the set of repelling periodic points: $x \in \mathbb{Z}_2$ with $f^n(x) = x$ for some $n \geq 1$ and $|(f^n)'(x)|_2 > 1$;
- (J4) the closure of the set (J3).

Under each reading we prove $J = \emptyset$. Hence (A) fails ($\dim_H \emptyset = 0$) while (B) holds ($\mathbb{Z}_2 \setminus J = \mathbb{Z}_2$), and the conjunction is false.

Conventions. (i) In (J3) the period $n \geq 1$ is any period, not only the minimal one; this makes the set (J3) larger, so emptiness under it implies emptiness for minimal periods. (ii) The multiplier $(f^n)'(x)$ is the derivative at $x \in \mathbb{Z}_2 \subset \mathbb{Q}_2$ of the n -th iterate of the same polynomial $g(z) = z^2 - 1$ on the field $\mathbb{Q}_2$ (Mathlib `deriv`), since $\mathbb{Z}_2$ is not a field. (iii) Hausdorff dimension is Mathlib’s `dimH` for the 2-adic metric on $\mathbb{Z}_2$; $\dim_H \emptyset = 0$ (`dimH_empty`). Under the other common convention $\dim_H \emptyset = -\infty$, the value is still not 1. (iv) Equicontinuity is Mathlib’s `EquicontinuousOn` / `EquicontinuousAt` for the uniform structure of the 2-adic metric. The chordal metric on affine points of $\mathbb{P}^1(\mathbb{Q}_2)$, $\rho(x, y) = |x - y|_2 / (\max(1, |x|_2) \max(1, |y|_2))$, equals $|x - y|_2$ on $\mathbb{Z}_2$ (Lean `chordal_eq_dist`), so it does not matter which of the two metrics is used on $\mathbb{Z}_2$.

2 Definitions and sources

The non-archimedean Fatou and Julia sets are defined by equicontinuity. Wikipedia, *Arithmetic dynamics* (https://en.wikipedia.org/wiki/Arithmetic_dynamics): “The metric on K and the standard definition of equicontinuity leads to the usual definition of the Fatou and Julia sets of a rational map”, and “A striking difference is that in the nonarchimedean setting, the Fatou set is always nonempty, but the Julia set may be empty.” Benedetto and Lee, *J-Stability in non-archimedean dynamics*, arXiv:2102.05841, Section 2: “The type I Fatou set consists of those points of $\mathbb{P}^1(\mathbb{C}_v)$ having a neighborhood on which the sequence of iterates is equicontinuous with respect to the chordal metric”, and a periodic point whose multiplier has absolute value greater than 1 “is said to be repelling”. Over $\mathbb{C}$, Wikipedia, *Julia set* (https://en.wikipedia.org/wiki/Julia_set): $J(f)$ “is the closure of the set of repelling periodic points.” Readings (J1) and (J2) follow the equicontinuity definition, (J3) the parenthetical “repelling points”, and (J4) the complex-dynamics characterization.

3 Proof

Lemma 1 (`lipschitz_f`, `lipschitz_iterate`). *For $x, y \in \mathbb{Z}_2$, $|f(x) - f(y)|_2 \leq |x - y|_2$. Hence every iterate f^n is 1-Lipschitz.*

Proof. $f(x) - f(y) = (x - y)(x + y)$ and $|x + y|_2 \leq 1$ in $\mathbb{Z}_2$. Iterate. $\square$

Theorem 2 (`fatouSet_eq_univ`, `juliaSet_eq_empty`, `juliaSetPt_eq_empty`). *The family $\{f^n\}_{n \in \mathbb{N}}$ is uniformly equicontinuous on $\mathbb{Z}_2$. Hence the Fatou set is all of $\mathbb{Z}_2$, and the sets (J1) and (J2) are empty.*

Proof. A family of functions that are all 1-Lipschitz is uniformly equicontinuous (Mathlib `LipschitzWith.uniformEquicontinuous`). Every point then has the neighbourhood $\mathbb{Z}_2$ on which the iterates are equicontinuous. $\square$

Lemma 3 (`hasDerivAt_iterate`, `multiplier_eq`). *For $z \in \mathbb{Q}_2$, $(g^n)'(z) = \prod_{i < n} 2g^i(z)$. For $x \in \mathbb{Z}_2$, $g^i(x) = f^i(x) \in \mathbb{Z}_2$, so $(g^n)'(x) = \prod_{i < n} 2f^i(x)$.*

Proof. Chain rule by induction on n , using $g^{n+1} = g \circ g^n$ and $g'(w) = 2w$. The coercion $\mathbb{Z}_2 \rightarrow \mathbb{Q}_2$ intertwines f and g . $\square$

Theorem 4 (`norm_multiplier_le`, `norm_multiplier_lt_one`, `repellingPeriodicPts_eq_empty`, `juliaRep_eq_empty`). *For every $x \in \mathbb{Z}_2$ and $n \geq 0$, $|(g^n)'(x)|_2 \leq 2^{-n}$; for $n \geq 1$ it is < 1 . Hence no point of $\mathbb{Z}_2$ is repelling (periodic or not), and the sets (J3) and (J4) are empty.*

Proof. $|2|_2 = 1/2$ and $|f^i(x)|_2 \leq 1$, so each factor has absolute value at most $1/2$. The closure of the empty set is empty. $\square$

Theorem 5 (`julia_facts`, `conjecture745_false`). *Under each reading (J1)–(J4), $J = \emptyset$, $\dim_H J = 0$, and $\mathbb{Z}_2 \setminus J = \mathbb{Z}_2$ is open and dense. Therefore, for each reading, “ $\dim_H J = 1$ and $\mathbb{Z}_2 \setminus J$ is open and dense” is false: conjunct (B) holds, conjunct (A) fails.*

Remark (not formalized). The same argument shows $\mathbb{Z}_2$ lies in the Fatou set of f on $\mathbb{P}^1(\mathbb{Q}_2)$ or $\mathbb{P}^1(\mathbb{C}_2)$: the closed unit disk is an open neighbourhood of each point of $\mathbb{Z}_2$ that f maps into itself, f is 1-Lipschitz there, and the chordal metric equals $|x - y|_2$ there. So $J \cap \mathbb{Z}_2 = \emptyset$ in those settings too. This remark is not part of the Lean development.

4 Lean formalization

File `lean/Conjecture745/Basic.lean` (Lean 4, Mathlib v4.33.1; only import `Mathlib`). Objects: `C745.f :  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$` (Mathlib `PadicInt 2`), `f x = x^2 - 1`; `C745.g` the same polynomial on $\mathbb{Q}_2$ (Mathlib `Padic 2`); `fatouSet`, `juliaSet` (J1), `juliaSetPt` (J2), `multiplier n x = deriv (g^[n]) x`, `repellingPeriodicPts` (J3), `juliaRep` (J4), `chordal`. Main theorem: (Unicode transliterated to ASCII: $\sim$ is $\neg$, $\wedge$ is $\wedge$, $\wedge^c$ is the set complement.)

```
theorem conjecture745_false :
  ~ (dimH juliaSet = 1 /\ IsOpen juliaSet^c /\ Dense juliaSet^c) /\
  ~ (dimH juliaSetPt = 1 /\ IsOpen juliaSetPt^c /\ Dense juliaSetPt^c) /\
  ~ (dimH repellingPeriodicPts = 1 /\ IsOpen repellingPeriodicPts^c /\
    Dense repellingPeriodicPts^c) /\
  ~ (dimH juliaRep = 1 /\ IsOpen juliaRep^c /\ Dense juliaRep^c)
```

`julia_facts` states that all four sets are empty, have `dimH 0`, and have open dense complements.

Axiom audit. `#print axioms` for `C745.conjecture745_false`, `C745.julia_facts` and `C745.chordal_eq_dist` reports only `[propext, Classical.choice, Quot.sound]`. No sorry, no `native_decide`, no added axioms. `lake build` succeeds.